We shall work with signals that are non-zero only for positive times $t > 0$.

Laplace transform of an analogue signal $x(t)$

$$X(s) = \mathcal{L}[x] = \int_0^\infty e^{-st} x(t) \mathrm{d}t \ , \qquad s \in \mathbb{C} \ , \quad X : \mathbb{C} \to \mathbb{C}$$

$s = \sigma + i\omega$ - **complex frequency**

Sampling the continuous signal:

$$\hat{x}(t) = x(t) \cdot \text{Ш}_T(t)$$

$$\mathcal{L}[\hat{x}] = T \sum_{n=0}^{\infty} \int_0^\infty x(t) \delta\left(t - nT\right) e^{-st} \mathrm{d}t = T \sum_{n=0}^{\infty} x[n] e^{-snT} = T \cdot X(z) \ , \quad z \equiv e^{sT}$$

z-transform of a digital signal $x[n]$

$$X(z) \equiv \sum_{n=0}^{\infty} x[n] z^{-n} \ , \qquad z \in \mathbb{C} \ , \quad X : \mathbb{C} \to \mathbb{C} \ .$$

$y(t) = h(t) * x(t) \ \rightarrow \ Y(s) = H(s) \cdot X(s) \qquad y[n] = h[n] * x[n] \ \rightarrow \ Y(z) = H(z) \cdot X(z)$

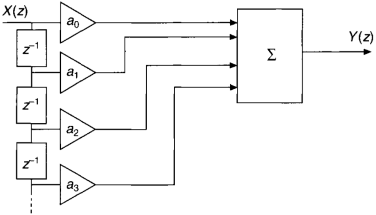
$H(s)$ and $H(z)$ transfer functions.

If $x(t) = \delta(t) \rightarrow X(s) = 1 \quad \rightarrow \quad Y(s) = H(s) \cdot 1$.

If $x[n] = \delta[n] \rightarrow X(z) = 1 \quad \rightarrow \quad Y(z) = H(z) \cdot 1$.

$\rightarrow$ transfer functions $H(s)$ and $H(z)$ are the transforms of the impulse response.

FIR filters and the z-domain



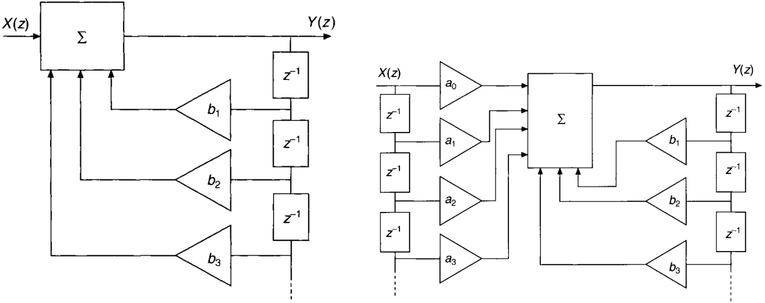
If

$$y[n] = a_0 x[n] + a_1 x[n-1] + \cdots = \sum_{l=0}^{k} a_l x[n-l]$$

$$Y(z) = \sum_{l=0}^{k} a_l z^{-l} X(z)$$

$$H(z) = \sum_{l=0}^{k} a_l z^{-l} = \frac{\sum_{l=0}^{k} a_l z^{k-l}}{z^k} = a_0 \frac{\prod_{l=1}^{k} (z - d_l)}{z^k}$$

IIR filters and the z-domain



If we have causal higher order LTI defined as

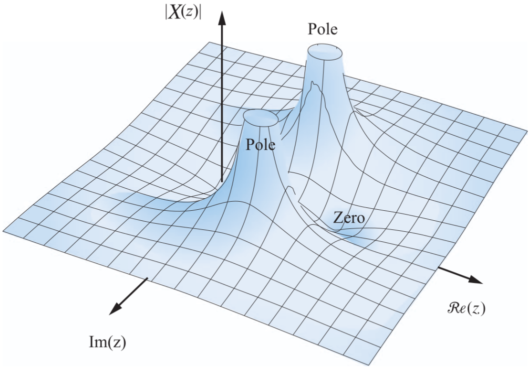
$$\sum_{l=0}^{m} b_l \cdot y[n-l] = \sum_{l=0}^{k} a_l \cdot x[n-l]$$

then

$$B(z) \cdot Y(z) = A(z) \cdot X(z) \quad \rightarrow \quad Y(z) = \frac{A(z)}{B(z)} X(z) \quad \rightarrow \quad H(z) = \frac{A(z)}{B(z)} .$$

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_k z^{-k}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_m z^{-m}} = \frac{z^m \sum_{l=0}^{k} a_l z^{k-l}}{z^k \sum_{l=0}^{m} b_l z^{m-l}} =$$
$$= \frac{a_0}{b_0} \cdot z^{m-k} \frac{\prod_{l=1}^{k}(z - d_l)}{\prod_{l=1}^{m}(z - c_l)} ,$$

where we assumed that $b_m \neq 0, a_k \neq 0$ and d_l are the zeros of the numerator and as such the **zeros** of $H(z)$ while c_l are the zeros of the denominator and as such the **poles** of $H(z)$.



- Recursive (IIR) filters can be used to achieve the same processing as almost all non-recursive (FIR) filters but have the advantage of needing fewer multipliers coefficients and so are simpler in terms of hardware.
- Because of the feedback, a recursive filter, i.e. IIR, can be unstable. FIR filters are stable.
- IIR filters also have a 'phase problem' compared to FIR filters.

$$\boxed{\text{continuous signal} \rightarrow s\text{ - plane}} \overset{\text{sampling}}{\Longrightarrow} \boxed{\text{digital signal} \rightarrow z\text{ - plane}}$$

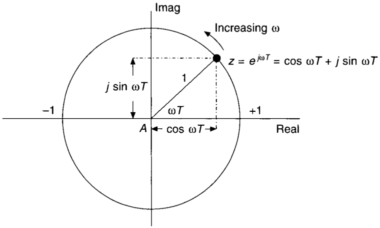
$$s = \sigma + i\omega \Longrightarrow z = e^{sT} = e^{\sigma T} e^{i\omega T} = |z| e^{i\omega T}$$

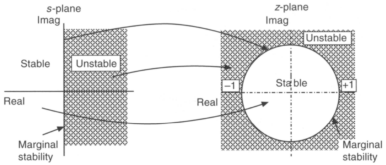
stability criterion

$$\sigma < 0 \Longrightarrow |z| < 1$$

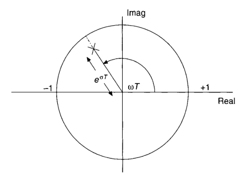
stability boundary

$$\sigma = 0 \Longrightarrow |z| = 1 \rightarrow z = e^{i\omega T}$$

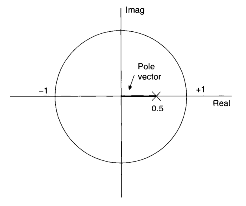


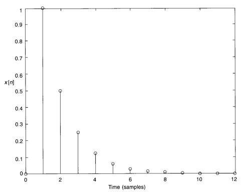


Example

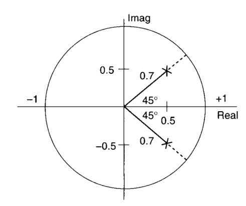


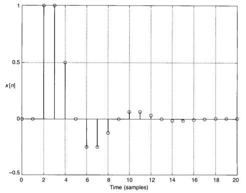
$T = 0.1$ s, $\sigma = -2$
$e^{\sigma T} \approx 0.82$





$$H(z) = \frac{1}{z - 0.5}$$





$$H(z) = \frac{1}{z^2 - z + 0.5}$$

Generally

$$\text{Gain} = k\frac{\prod \text{zero distances}}{\prod \text{pole distances}}$$

$$\text{Phase angle} = \sum \text{zero angles} - \sum \text{pole angles}$$

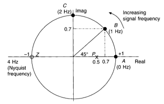
k is the **pure gain** of the system.
If there is no zero then one must be used for the zero distances.

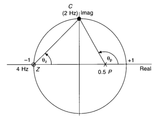
The frequency point moves around the unit circle in an anticlockwise direction as the signal frequency increases.
The point starts at $z = +1$ (d.c.), and ends at $z =$ -1 (the Nyquist frequency).

Example

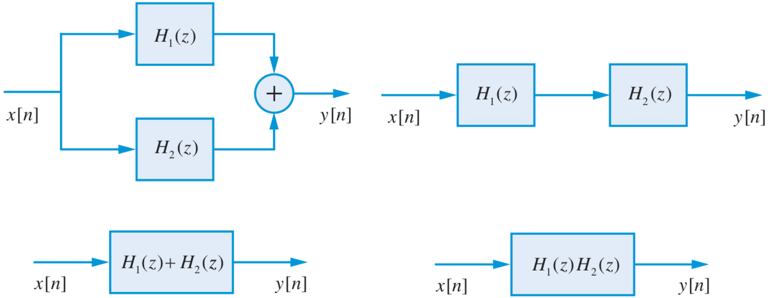
Find the response at 0 Hz (d.c), 1 Hz and 2 Hz for

$$H(z) = \frac{z+1}{z-0.5} \quad \text{if} \quad f_s = 8 \text{ Hz},$$

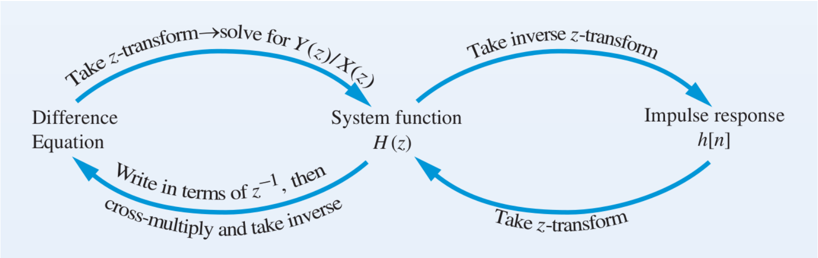




$f = 0$ Hz $\rightarrow$ Gain $= 4$, phase angle $= 0^{\circ}$
$f = 1$ Hz $\rightarrow$ Gain $=1.84/0.73 \approx 2.5$, phase angle $= \arctan(0.7/1.7) - \arctan(0.7/0.2) = -51.8^{\circ}$
$f = 2$ Hz $\rightarrow$ Gain $= \frac{\sqrt{2}}{\sqrt{1.25}} = 1.27$, phase angle $= 45^{\circ} - 116.5^{\circ} = -71.5^{\circ}$



(a) (b)



Take z-transform→solve for $Y(z)/X(z)$
Take inverse z-transform
Difference
Equation
System function
$H(z)$
Impulse response
$h[n]$
Write in terms of z^{-1}, then
cross-multiply and take inverse
Take z-transform